Notes

Prevalence-calibrated density-ratio weighting

Estimand, identification, and evidence in `i3pw`

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`https://github.com/bvilhjal/i3pw` · v0.3.1

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Abstract

Volunteer biobanks and ascertained genetic cohorts need not have the same distribution as their source populations. `i3pw` combines a base weight from a participation model with an exponential tilt chosen to reproduce externally known disease prevalences. In the one-disease case this is the classical choice-based correction: cases receive relative weight K/P and controls $(1-K)/(1-P)$. In general it is an entropy-calibration estimator. The weights equal full-population inverse-probability weights only when the population-to-participant density ratio lies in the span of the base plus the chosen moments. Matching a register prevalence is an identity. Held-out margins can refute the tilt; they do not prove it. A benchmark suite over six axes of that specification puts numbers on the consequences: the two ingredients fail completely in each other's regime, a participation model used as a base introduces bias when recruitment acts through the diagnosis alone, and interval coverage falls from nominal to 0.64 under the mild misspecification of ordinary applied use. Numeric claims below are copied from two fixed-seed artifacts, `report/validation_results.tsv` (i3pw 0.3.0) and `report/benchmark_results.tsv` (i3pw 0.3.1); the tables and figures reporting the latter are generated from it. None of it is external-cohort validation.

The intended reader already uses GWAS, case–control ascertainment, and the Lee liability transform. These notes do not re-derive those. They fix the estimand, the identifying density-ratio family, what λ is, which genetic analyses the weights can repair, and which simulation claims survive their design. The import name `i3pw` is historical. The accurate name for the estimator is prevalence-calibrated density-ratio weighting.

Contents

1 Estimand

A register margin is a moment constraint. Among participant weights satisfying that constraint, `i3pw` chooses the weights closest in Kullback–Leibler divergence to a user-supplied base [6, 11, 7, 18, 15, 13]. Table 1 is the shortest accurate description.

Table 1: The shortest accurate description of `i3pw`.

Question	Answer
What is computed?	Normalized minimum-divergence weights matching specified population moments.
What information is added?	External prevalences or other population margins, optionally layered on weights from a participation model.
What is identified?	The target population law if its density ratio relative to participants is the base ratio times the chosen exponential tilt; otherwise an information projection, not the true inclusion probabilities.
What is λ?	The fitted coefficient in the log population-to-participant density ratio. It is not, in general, the coefficient of outcome in a participation logit.
Which IP weight targets the full population?	$1/\pi$, where $\pi = \mathbb{P}(S = 1 \mid X, Y)$. The inverse odds $(1-\pi)/\pi$ target nonparticipants.
Does matching prevalence validate the method?	No. A held-out margin can diagnose failure, but its standardized mean difference is a heuristic discrepancy, not a formal Sargan–Hansen J-test.
How strong is the present evidence?	Theoretical links, unit tests, the 0.3.0 example freeze in `report/validation_results.tsv`, and the 0.3.1 benchmark suite in `report/benchmark_results.tsv`, which measures held-out bias, coverage and failure rates against oracle weights across six axes of the specification. All of it is simulation; there is no reported external-cohort validation.

1.1 The problem in a geneticist's language

Let $S = 1$ denote participation, X variables available on the sampling frame, and Y one or more diagnoses observed in participants. Let T denote the source population to which inference is intended to apply. A volunteer cohort usually samples from $\mathbb{P}(X, Y \mid S = 1)$, whereas the scientific target is a functional of $\mathbb{P}_T(X, Y)$. This is a distribution-shift problem.

For a geneticist, the simplest instance is ordinary case–control ascertainment. If cases constitute P of the analysed sample but K of the population, a population mean is recovered by

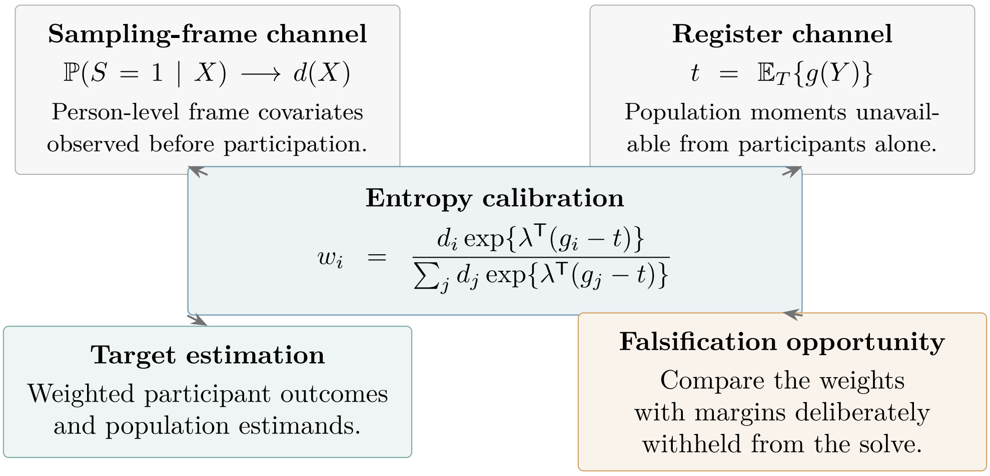

Figure 1: The two information channels and two uses of prevalence-calibrated weighting. The participation model supplies individual variation visible in frame covariates; the register anchors selected population moments. Target estimates use the resulting weights, whereas held-out margins provide a diagnostic that can fail.

giving cases relative weight K/P and controls relative weight $(1-K)/(1-P)$. `i3pw` keeps that calculation but permits:

1. a nonuniform starting weight $d(X)$;
2. several prevalence, comorbidity, or stratum-specific constraints; and
3. downstream weighted means and liability-scale calculations.

The price of this generality is that one must say exactly what selection model these constraints identify.

1.2 A small epidemiologic dictionary

2 Probability setup: what full-population IPW means

Write $Z = (X, Y)$ and

$$\pi(z) = \mathbb{P}(S = 1 \mid Z = z), \qquad p_S = \mathbb{P}(S = 1). \tag{1}$$

Bayes' rule gives the population-to-participant density ratio

$$r_T(z) = \frac{d\mathbb{P}_T}{d\mathbb{P}_1}(z) = \frac{p_S}{\pi(z)}, \qquad \mathbb{P}_1(\,\cdot\,) = \mathbb{P}_T(\,\cdot \mid S = 1). \tag{2}$$

Consequently, for any integrable target variable $h(Z)$,

$$\mu_T = \mathbb{E}_T\{h(Z)\} = \frac{\mathbb{E}_1\{h(Z)/\pi(Z)\}}{\mathbb{E}_1\{1/\pi(Z)\}}. \tag{3}$$

Equation (3) is the self-normalized, or Hájek, form of Horvitz–Thompson weighting [10, 8]. The unknown sampling fraction p_S cancels. It does not follow that $\pi(z)$ can be learned from participant data and one marginal prevalence: that is the identification problem addressed below.

Table 2: Terms used in this report and their statistical-genetic analogues.

Term	Meaning here	Familiar analogue
Target population	The people about whom the estimand is defined, including nonparticipants.	The population whose disease prevalence K enters a liability transform.
Participation probability	$\pi(x, y) = \mathbb{P}(S = 1 \mid X = x, Y = y)$.	An ascertainment or recruitment probability.
Positivity	Every target-relevant type has nonzero chance to appear.	Both cases and controls, and every required stratum, must be represented.
Calibration	Choose weights so specified weighted sample moments equal known population moments.	Force the weighted case fraction to equal K.
Density ratio	$d\mathbb{P}_T/d\mathbb{P}(\cdot \mid S = 1)$, the weight that transports one law to another.	A sample-to-population prior or ascertainment correction.
Collider bias	An association induced or distorted by conditioning on a common effect such as participation.	Genotype and phenotype both influence volunteering, so analysing volunteers can associate them spuriously [16, 23].

2.1 Inverse probability is not inverse odds

There are two legitimate transports, but they have different destinations. From Bayes' rule,

$$\frac{d\mathbb{P}_T}{d\mathbb{P}_1}(z) \propto \frac{1}{\pi(z)}, \tag{4}$$

$$\frac{d\mathbb{P}_0}{d\mathbb{P}_1}(z) \propto \frac{1-\pi(z)}{\pi(z)}, \qquad \mathbb{P}_0(\,\cdot\,) = \mathbb{P}_T(\,\cdot \mid S = 0). \tag{5}$$

Thus $1/\pi$ targets the full source population, whereas inverse odds target nonparticipants. Logistic modelling does not make the two targets identical. When participation is rare, $1 - \pi(z) \approx 1$, so the normalized weights can be close; this is an approximation, not an algebraic equivalence.

3 The one-outcome calculation

Let $Y \in \{0, 1\}$, let

$$P = \mathbb{P}(Y = 1 \mid S = 1), \qquad K = \mathbb{P}_T(Y = 1), \tag{6}$$

and begin from uniform base weights. An exponential tilt gives cases relative weight $\exp(\lambda)$ and controls weight one. Requiring the weighted case fraction to equal K yields

$$\frac{P\exp(\lambda)}{P\exp(\lambda) + (1-P)} = K \quad\Longrightarrow\quad \lambda = \operatorname{logit}(K) - \operatorname{logit}(P). \tag{7}$$

After an irrelevant common normalization, the class weights are

$$d_1 = \frac{K}{P}, \qquad d_0 = \frac{1-K}{1-P}. \tag{8}$$

For $P = 0.20$ and $K = 0.40$, equation (7) gives $\lambda = 0.9808$; the relative weights are 2.00 for a case and 0.75 for a control. This is worth checking by hand before trusting a larger solve. This is the classical choice-based-sampling result [15] and the core ascertainment factor appearing in liability-scale corrections [13].

The interpretation of λ matters. Let $\pi_y = \mathbb{P}(S = 1 \mid Y = y)$. Equations (2) and (7) imply

$$\lambda = \log\left(\frac{\pi_0}{\pi_1}\right). \tag{9}$$

It is therefore a coefficient in the *population-to-participant density ratio*. Suppose instead that participation obeys $\operatorname{logit}(\pi_y) = \alpha + \gamma y$. Then

$$\lambda = -\gamma + \log\left\{\frac{1 + \exp(\alpha + \gamma)}{1 + \exp(\alpha)}\right\} = -\gamma + \log\left(\frac{1 - \pi_0}{1 - \pi_1}\right). \tag{10}$$

Only under rare participation is the second term negligible, giving $\lambda \approx -\gamma$. Calling λ the participation-logit coefficient would therefore reverse a sign and omit a non-negligible term in many cohort settings.

4 General entropy calibration

Let $g_i = g(Y_i, X_i) \in \mathbb{R}^k$ be calibration features, with externally known target

$$t = \mathbb{E}_T\{g(Y, X)\}. \tag{11}$$

Examples include diagnosis indicators, pairwise comorbidity indicators, stratum shares, and diagnosis-by-stratum products. Let $d_i \geq 0$ be normalized base weights. Uniform d_i give pure prevalence calibration; estimated $1/\widehat{\pi}(X_i)$ supply a covariate correction first.

4.1 Primal and dual problems

For exact calibration, `i3pw` solves the empirical information projection

$$\begin{aligned} &\min_{w_1,\dots,w_n} \quad \sum_{i=1}^{n} w_i \log\left(\frac{w_i}{d_i}\right) \\ &\text{subject to} \quad w_i \geq 0, \qquad \sum_{i=1}^{n} w_i = 1, \qquad \sum_{i=1}^{n} w_i g_i = t. \end{aligned} \tag{12}$$

This is a Kullback–Leibler calibration estimator [4, 6, 7]. Its dual is

$$\widehat{\lambda} = \arg\min_{\lambda \in \mathbb{R}^k} \log\left[\sum_{i:d_i>0} d_i \exp\{\lambda^\mathsf{T}(g_i - t)\}\right], \tag{13}$$

and the resulting weights are

$$w_i(\lambda) = \frac{d_i \exp\{\lambda^\mathsf{T}(g_i - t)\}}{\sum_{j:d_j>0} d_j \exp\{\lambda^\mathsf{T}(g_j - t)\}}. \tag{14}$$

Rows with $d_i = 0$ remain outside the support and receive zero weight. The optimization is only k-dimensional even when n is large.

Empirical likelihood and entropy balancing are close relatives, not the same optimization. Both impose moment restrictions and belong to generalized empirical likelihood families. Standard empirical likelihood maximizes $\sum_i \log w_i$ and produces reciprocal-affine weights; the criterion in (12) produces exponential weights. Results from one should not be attributed mechanically to the other [19, 18].

4.2 Ridge stabilization

With ridge parameter $\rho > 0$, the implemented dual becomes

$$\widehat{\lambda}_\rho = \arg\min_{\lambda} \left[\log \sum_{i:d_i>0} d_i \exp\{\lambda^\mathsf{T}(g_i - t)\} + \frac{\rho}{2}\|\lambda\|_2^2 \right]. \tag{15}$$

Its gradient and Hessian are

$$\nabla \ell_\rho(\lambda) = \mathbb{E}_{w(\lambda)}(g) - t + \rho\lambda, \tag{16}$$

$$\nabla^2 \ell_\rho(\lambda) = \mathrm{Cov}_{w(\lambda)}(g) + \rho I_k. \tag{17}$$

At the solution,

$$\mathbb{E}_{w(\widehat{\lambda}_\rho)}(g) - t = -\rho\widehat{\lambda}_\rho. \tag{18}$$

Ridge calibration is therefore deliberately approximate. It trades bias in the specified margins against less concentrated weights; it must not be reported as exact calibration.

4.3 Feasibility and positivity

For a finite exact solution, t must lie in the relative interior of the convex hull of the observed g_i having positive base weight. A binary target strictly between zero and one is unreachable if the sample contains only cases or only controls. Near a hull boundary, $\|\widehat{\lambda}\|$ and the largest weight can explode even when the optimizer reports success. Redundant constraints can also make λ nonunique. These are data-support problems, not numerical inconveniences that a different optimizer can wish away.

5 Identification: when calibration is population weighting

The empirical solve always defines weights if its constraints are feasible. The scientific question is when those weights recover the target law.

Density-ratio assumption. Suppose the true population-to-participant ratio can be represented as

$$r_T(X, Y) = c\, d(X) \exp\{\theta^\mathsf{T} g(X, Y)\}, \tag{19}$$

where c normalizes the ratio. Here $d(X)$ is the population-to-participant correction already supplied by the base model, and g spans the remaining selection pattern. If the target moments are correct, positivity holds, and the moment map is identifiable, then the population moment equation

$$\frac{\mathbb{E}_1[d(X)\exp\{\theta^\mathsf{T} g\}g]}{\mathbb{E}_1[d(X)\exp\{\theta^\mathsf{T} g\}]} = t \tag{20}$$

identifies θ; the sample solution estimates it. Under (19), the calibrated weights are proportional to the true full-population IP weights.

This statement is the right meaning of *prevalence informs the weights.* It does not say that prevalence nonparametrically identifies $\pi(X, Y)$. Rather, prevalence identifies the low-dimensional coefficient of a density-ratio family chosen in advance.

5.1 What happens under misspecification

If equation (19) is false, the solve still returns the member of the exponential family that matches t and is closest to the base distribution in KL divergence. Let r_λ denote that fitted ratio. For a downstream variable h, its transport bias is

$$\mathbb{E}_1[\{r_\lambda(Z) - r_T(Z)\}h(Z)]\,. \tag{21}$$

Calibration gives no general reason for equation (21) to vanish. It does vanish for functions in the affine span of the constrained moments:

$$h(Z) = a + b^\mathsf{T} g(Z) \quad \Longrightarrow \quad \mathbb{E}_{w_\lambda}(h) = a + b^\mathsf{T} t. \tag{22}$$

This explains both the method's strength and its limit. Anchored quantities are right by construction; transfer to unanchored quantities depends on how well the chosen g captures the relevant distribution shift.

5.2 Label shift and case mix

With uniform base weights, a categorical Y, and $g(Y)$ containing a full set of class indicators, the assumption reduces to

$$\mathbb{P}_T(X \mid Y) = \mathbb{P}_1(X \mid Y). \tag{23}$$

That is, only class proportions change. This is prior-probability or label shift [21, 14]. It is plausible for deliberately sampled case–control data when recruitment is random within case status. It is doubtful when, for example, young severe cases enter at a different rate from old mild cases.

For Q comorbid binary diagnoses, matching only the Q marginal prevalences is not full label-shift correction over the 2^Q joint disease patterns. If participation treats a comorbid case differently from the product of its marginal diagnoses, the corresponding joint prevalence or interaction must be constrained.

A known $K = \mathbb{P}_T(Y = 1)$ sets the number of cases, not their composition: $\mathbb{P}_T(X \mid Y = 1)$ remains unidentified. If population prevalences are available within sex, birth cohort, ancestry, region, or severity strata, interactions such as

$$g_{aq}(X, Y) = \mathbf{1}(A = a)Y_q \tag{24}$$

can represent that structure. The number of constraints then grows quickly, so richer identification consumes positivity and effective sample size. There is no free lunch; the convex hull sends an invoice.

6 Target estimands in statistical genetics

Weighting changes a probability law. Whether that repairs a scientific analysis depends on whether the analysis is a functional of that law and whether the fitted density ratio is correct.

6.1 Participation as a collider

Suppose genotype G or an exposure E influences participation and disease Y also influences participation. Conditioning on volunteers opens a path between the causes of S, potentially

Table 3: What prevalence calibration can and cannot establish for common genetic analyses.

Estimand	When the weighting helps	Principal caveat
Disease prevalence	An exact anchor reports the supplied population prevalence.	This is not an empirical result; uncertainty in the external prevalence remains.
Population mean or absolute risk	Consistent if the fitted ratio transports the participant distribution of the measured variable to the target population.	Marginal disease calibration alone does not fix within-disease selection on severity or prognosis.
Allele frequency or biomarker mean	Can transfer when its relation to the constrained features is stable, or when the full density-ratio model is correct.	A known disease prevalence contains no direct information about genotype or biomarker composition within disease classes.
Logistic GWAS coefficient	Under selection depending on case status alone, the odds-ratio slope is invariant to case–control sampling [17].	When genotype or its causes and phenotype both affect participation, conditioning on $S = 1$ can create collider bias; a marginal K does not identify the required joint correction [16, 23].
Liability-scale R^2 or heritability	Case/control weights recover the population case fraction before the observed-to-liability transform; this parallels classical ascertainment correction [5, 13].	The result inherits the liability-threshold model and the assumed selection law. The current pairwise implementation is quadratic in sample size.
Causal effect	Weights can address selection into the analysed sample.	They do not by themselves address exposure confounding, measurement error, treatment assignment, or interference.

distorting G–Y or E–Y associations. This is not repaired merely by restoring $\mathbb{P}(Y = 1) = K$, because infinitely many joint laws $\mathbb{P}_T(G, Y)$ share the same marginal K. Volunteer-weighting studies in UK Biobank illustrate both the importance and the difficulty of modelling this selection [23, 24, 25].

The distinction from Prentice–Pyke invariance is useful:

- if S depends only on Y, a correctly specified logistic disease slope can remain consistent without prevalence weighting; and
- if S depends jointly on G, X, and Y, population associations require a correction rich enough for that joint selection.

Prevalence calibration can be one component of that correction, but it cannot replace the other components.

6.2 Liability-scale quantities

For population prevalence K, threshold $t_K = \Phi^{-1}(1 - K)$, and $z_K = \phi(t_K)$, a common observed-to-liability conversion is

$$R_L^2 = R_O^2 \frac{K(1-K)}{z_K^2}. \tag{25}$$

If R_O^2 was estimated in a case–control sample with case fraction P, the Lee et al. ascertainment factor gives

$$R_L^2 = R_{O,\text{asc}}^2 \times \frac{K(1-K)}{z_K^2} \times \frac{K(1-K)}{P(1-P)}. \tag{26}$$

Alternatively, one may first use equation (8) to estimate the population-standardized observed-scale moment and then apply equation (25). Their agreement is expected under pure case/control ascertainment. Neither route protects against recruitment depending on latent liability within case status unless that dependence is modelled.

7 Uncertainty: what is random?

There are at least four uncertainty sources: sampling participants, estimating the base participation model, fitting the tilt, and estimating or transporting the external targets. A standard error that conditions on all weights and targets treats only the first.

7.1 Fixed-weight and calibration-aware linearization

For normalized weights $\widetilde{w}_i = w_i / \sum_j w_j$, the fixed-weight variance of $\widehat{\mu} = \sum_i \widetilde{w}_i h_i$ is estimated by

$$\widehat{\mathrm{Var}}_{\mathrm{fixed}}(\widehat{\mu}) = \sum_{i=1}^{n} \widetilde{w}_i^2 (h_i - \widehat{\mu})^2. \tag{27}$$

This ignores that the tilt was re-estimated to meet the constraints. The calibration-aware linearization implemented in `i3pw` instead uses

$$\widehat{\beta}_\rho = \{\widehat{\mathrm{Cov}}_w(g) + \rho I\}^{-1} \widehat{\mathrm{Cov}}_w(g, h), \tag{28}$$

$$e_i = h_i - \widehat{\mu} - \widehat{\beta}_\rho^{\mathsf{T}} (g_i - \bar{g}_w), \tag{29}$$

$$\widehat{\mathrm{Var}}_{\mathrm{cal}}(\widehat{\mu}) = \sum_{i=1}^{n} \widetilde{w}_i^2 e_i^2. \tag{30}$$

The software computes equation (28) by penalized least squares rather than explicitly inverting a possibly ill-conditioned covariance matrix.

If $\rho = 0$ and h is an exactly constrained column, the residual in equation (29) is zero up to numerical tolerance. Its conditional sampling standard error is consequently zero: the estimator always prints the fixed target. For $\rho > 0$, equation (18) shows that the target is not fixed, $\widehat{\beta}_\rho$ is shrunk, and the anchored margin retains uncertainty. Using the exact-calibration projection after a ridge solve would be wrong.

7.2 External targets, fitted bases, and dependence

Zero conditional standard error is not zero scientific uncertainty. If a registry target $\widehat{t}$ has covariance V_t, a first-order combination for an independent external source is

$$\mathrm{Var}(\widehat{\mu}) \approx \mathrm{Var}_{\mathrm{participants}}(\widehat{\mu} \mid \widehat{t}) + D_t V_t D_t^{\mathsf{T}}, \qquad D_t = \frac{\partial \mu}{\partial t^{\mathsf{T}}}. \tag{31}$$

Temporal mismatch, outcome-definition mismatch, ancestry mismatch, and overlap between registry and cohort create additional bias or covariance that equation (31) does not solve.

The package also offers resampling for its simulation wrapper. A scientifically complete bootstrap should re-fit every estimated stage that would change under repeated sampling, including the base model when applicable. Rare anchors can make bootstrap replicates infeasible; silently discarding precisely those replicates truncates the distribution and narrows intervals.

Family or household relatedness requires cluster-level resampling or a cluster-robust influence calculation. The current simple mean interface does not automate those genetic-cohort complications.

7.3 Augmented IPW

For a participant-only trait V, covariates X available across the population frame, and regression $m(X) = \mathbb{E}(V \mid X)$, the package implements

$$\widehat{\mu}_{\mathrm{AIPW}} = \frac{1}{N}\sum_{i=1}^{N}\widehat{m}(X_i) + \sum_{i:S_i=1}\widetilde{w}_i\{V_i - \widehat{m}(X_i)\}. \tag{32}$$

In the standard missing-data or data-integration model, this estimator can be consistent when either the outcome regression or the participation weighting is correct [20, 2]; flexible regressions call for sample splitting or cross-fitting [3]. This is not a blanket double-robustness theorem for prevalence calibration. AIPW is a *downstream* estimator for a participant-only trait when X is observed on the frame. It does not repair a misspecified tilt family: if $g(Y)$ misses the outcome-driven part of selection, the weighting piece and the outcome-regression piece can fail together. Similarly, the result that entropy balancing is doubly robust in particular causal models [26] does not prove that arbitrary outcome-dependent volunteer selection is repaired by a few prevalence margins.

8 Diagnostics and falsifiability

8.1 Quantities that diagnose the solve

Table 4: Diagnostics, their interpretation, and their limits.

Diagnostic	What it reveals	What it cannot reveal
Optimizer convergence	Whether the numerical dual solve terminated satisfactorily.	Whether the density-ratio family is scientifically correct.
Calibration residual	Whether exact targets were reached, or how far ridge calibration missed them.	Validity: an exact residual is expected by construction.
Kish effective sample size [12]	$\mathrm{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$, a concise measure of concentration.	A universal variance or information measure for every downstream analysis.
Weight range and top-1% mass	Dependence on a small number of observations.	Bias from omitted selection variables.
Support counts	Whether each required class or stratum is represented.	Adequate overlap within unmeasured subgroups.
Held-out SMD	Whether a known margin excluded from the solve is reproduced after weighting.	A formal hypothesis test, or proof that all unobserved margins are balanced.

For a held-out variable a with known population mean t_a, `i3pw` reports

$$\mathrm{SMD}_a = \frac{\sum_i \widetilde{w}_i a_i - t_a}{s_{a,\mathrm{sample}}}. \tag{33}$$

The often-used rule $|\mathrm{SMD}| < 0.1$ is a descriptive convention [1]. A held-out failure is valuable: it refutes the proposition that the fitted weighting transports that margin. A small SMD is only partial positive evidence.

Calling equation (33) a Sargan–Hansen J-test would overstate it [22, 9]. A formal overidentification test needs a joint covariance estimate, a quadratic test statistic, degrees of freedom, and an asymptotic reference distribution while accounting for estimated parameters. The present diagnostic borrows the *logic* of unused restrictions, not that formal test.

8.2 An honest validation design

Known margins should be divided before fitting:

1. use a prespecified subset to calibrate;
2. keep other registry quantities genuinely held out;
3. fit the tilt on one wave or fold and use the stored tilt on a new wave when possible; and
4. evaluate downstream quantities not algebraically determined by the anchors.

Selecting constraints after inspecting held-out failures converts validation into training. Reporting only anchored prevalence error produces impressive zeros and no evidence whatsoever.

9 Simulation evidence

The mathematical ingredients of `i3pw` are well established. What is new is their particular packaging and proposed use of register prevalences to augment participation weights. Evidence for that applied proposal must be judged separately from evidence for entropy calibration itself.

9.1 The 0.3.0 example freeze

Every number in Table 6 and Figure 2 is copied from `report/validation_results.tsv`. That freeze was produced on 11 August 2026 with i3pw 0.3.0, Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, and scikit-learn 1.9.0; the runtime is recorded in `report/validation_environment.txt`. The present package version is 0.3.1; the numeric artifact was not regenerated. Replicated scripts use seeds $0, \dots, R-1$. These are regression checks, not a substitute for an independently designed validation study.

Table 5: Current evidence and what each layer can support.

Evidence layer	What it supports	What it does not support
Established literature	IPW, survey calibration, information projection, density-ratio tilting, and case–control ascertainment correction.	That the selected g spans selection in a particular biobank.
Unit tests	Array contracts, numerical convergence, known special cases, support checks, and selected uncertainty calculations.	External validity, realistic misspecification, or calibrated interval coverage in a real cohort.
Synthetic data generator	Controlled comparison when outcomes and Bernoulli-logistic participation are known.	Performance under an unknown real recruitment process.
Example scripts and frozen local table	Illustrations of pure outcome selection, covariate-plus-outcome selection, collider settings, comorbidity, and liability models; selected corrected runs are recorded in `report/validation_results.tsv`.	Prespecified, independently replicated benchmark evidence.
Benchmark suite (§9.2)	Behaviour on held-out estimands as the recruitment law, the register information, the target error, the case mix, the support and the ridge are varied one at a time, each against oracle weights; and interval coverage under correct and incorrect specification.	That the simulated recruitment laws resemble a biobank's, or that any of it transfers outside the one data-generating family in `benchmarks/simulate.py`.
Real-cohort study	None is included in the present evidence base.	No empirical claim about bias reduction in iPSYCH, UK Biobank, or another cohort follows from the repository alone.

Table 6: Selected post-correction synthetic results. Values are mean ± standard deviation across repetitions unless stated otherwise.

Experiment	Quantity	Result	Interpretation
`honest_benchmark`, no X channel, $R = 12$	Absolute percentage error on an unanchored outcome	Naive 80.78 ± 6.66; calibrated LASSO base 81.61 ± 6.03; calibrated uniform base 82.24 ± 5.96.	Anchoring one disease did not transfer to another.
`honest_benchmark`, X channel strength 1.5, $R = 12$	Worst held-out |SMD|; absolute error in held-out X_0 mean	LASSO base 0.209 ± 0.079, 0.126 ± 0.072; uniform base 0.482 ± 0.238, 0.317 ± 0.211.	When a covariate channel exists, the base participation model materially helps.
`doubly_robust_trait`, $R = 20$	Mean bias; mean absolute bias for a participant-only trait	Naive $-0.0905, 0.1008$; LASSO IPW $-0.0252, 0.0636$; calibration $0.0537, 0.0864$; AIPW $-0.0009, 0.0584$.	Augmentation performed best in this particular, outcome-model-friendly design.
`monte_carlo`, $R = 20$	Anchored prevalence error; calibration ESS	Calibration error 0.00 ± 0.00; ESS 153 ± 19.	The zero error is the constraint, not validation; ESS quantifies its variance cost.

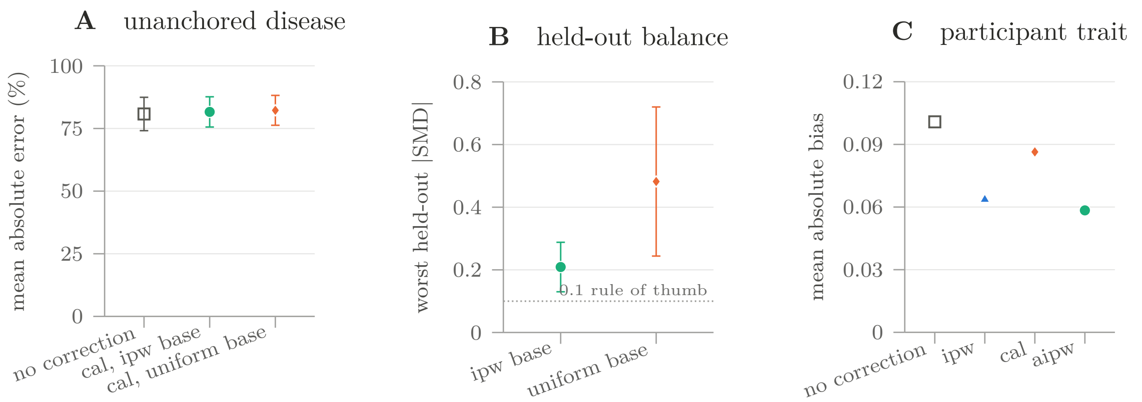

Figure 2: Three fixed-seed synthetic contrasts from the frozen validation artifact. Panel A shows mean absolute percentage error for an outcome excluded from calibration when selection has no direct covariate channel ($R = 12$). Panel B compares base weights when covariates do affect selection ($R = 12$); the dashed 0.1 line is only an SMD rule of thumb, not a hypothesis test. Panel C shows mean absolute bias for a participant-only trait in one outcome-model-friendly design ($R = 20$); it is not a general method ranking. Whiskers in A and B are ±1 across-seed standard deviation. The frozen artifact does not retain an uncertainty summary for the mean absolute biases in C.

The table and figure support the package's deliberately narrow messages: exact anchors are tautologies, cross-outcome transfer is weak, and the covariate base matters when the data-generating process actually contains a covariate selection channel. They do not establish bias correction in a real cohort, interval coverage, or robustness to an unknown selection law. Earlier editions of this note listed what stronger evidence would require; §9.2 is the attempt to supply it, and §9.2.8 states what is still missing afterwards.

9.2 The 0.3.1 benchmark suite

The freeze above illustrates the estimator. It does not test it: every experiment in it uses one recruitment mechanism, and the only estimand that can fail is a prevalence. The suite in `benchmarks/` was written against the validation matrix of §9.3 instead, one benchmark per axis, and it reports quantities the estimators were never given.

Its numbers are the artifact `report/benchmark_results.tsv`, produced by `python -m benchmarks.run_all` with the environment recorded in `report/benchmark_environment.txt`. Every seed is fixed, so the run reproduces exactly. The tables and figures that follow are *generated* from that artifact by `benchmarks/make_figures.py`: no number in this section was retyped, and a figure that could not find its row would have stopped the build rather than leave a gap.

The common design. A population of 20,000 carries eight correlated Gaussian frame covariates, a three-level stratum whose prevalence and participation both differ, a liability model giving a disease of prevalence $K = 0.10$ with a recorded severity grade, a second disease at 0.05, and a continuous trait Z. Roughly 2,500 people participate under a Bernoulli-logistic law whose channels are switched on one at a time. Forty replications, and three hundred where a coverage rate is being estimated.

Three choices make the results readable. First, the headline estimand is $\mathbb{E}[Z]$: a trait observed only in participants, never a calibration constraint, and correlated with both selection channels. It is a quantity the estimator can get wrong, which the anchored prevalence is not. Second, a constrained moment is never plotted, and where one appears in a table it is marked †. Third, every comparison carries an *oracle* row — weights $1/\pi$ built from the true inclusion probabilities. The oracle is not a competitor; it is the Monte Carlo floor at this sample size. Its distance from zero is irreducible noise, and an estimator's distance from the oracle is misspecification.

Table 7: B1. Held-out trait mean under five recruitment laws, 40 replications. `ipw` is a LASSO participation model on the frame; `cal` is calibration to the register prevalence from a uniform base; `ipw+cal` combines them; `aipw` augments `ipw+cal` with a cross-fitted outcome regression. The oracle column is the Monte Carlo floor, not a method.

Recruitment law	naive	ipw	cal	ipw+cal	aipw	oracle
Bias (population SD); signed						
X only	+0.389	+0.002	+0.383	+0.003	-0.002	-0.002
Y only	+0.301	+0.150	+0.006	-0.065	+0.046	+0.006
X + Y	+0.479	+0.120	+0.292	-0.036	+0.029	+0.002
X x Y	+0.558	+0.075	+0.310	-0.083	+0.033	+0.002
severity	+0.547	+0.158	+0.298	-0.040	+0.050	+0.002
RMSE (population SD)						
X only	0.393	0.024	0.384	0.025	0.020	0.024
Y only	0.304	0.153	0.022	0.083	0.048	0.023
X + Y	0.485	0.124	0.293	0.057	0.034	0.023
X x Y	0.567	0.085	0.311	0.106	0.039	0.023
severity	0.553	0.162	0.299	0.068	0.053	0.025
Kish effective sample size, of ∼2500 participants						
X only	2511	1370	2493	1371	—	1356
Y only	2509	2224	1906	1733	—	1907
X + Y	2511	1755	2147	1517	—	1543
X x Y	2513	1451	2072	1238	—	1411
severity	2512	1701	1982	1366	—	1415

9.2.1 Recruitment mechanism (B1)

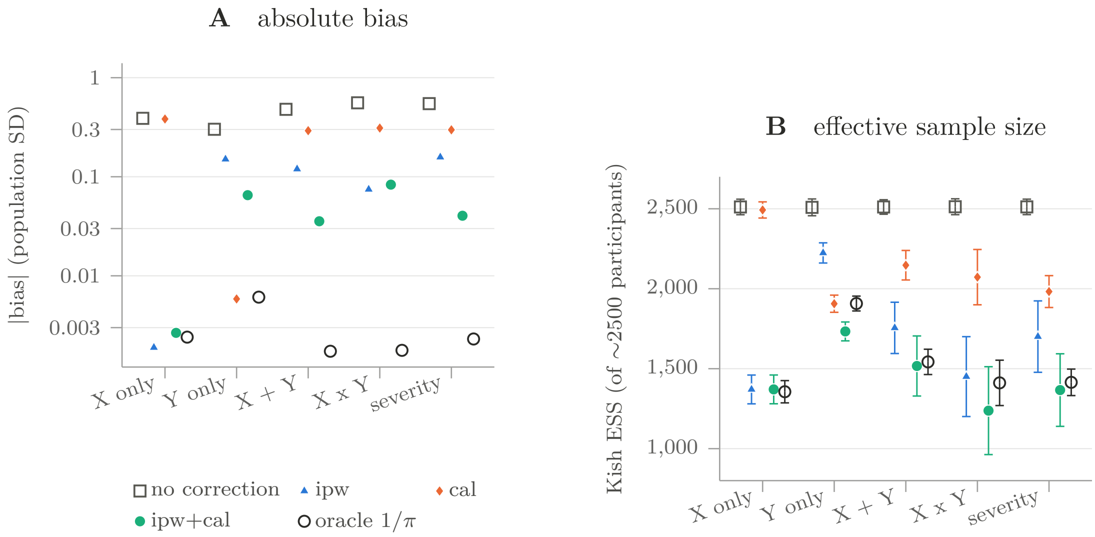

Figure 3: B1. Panel A is absolute bias on a logarithmic axis, because the estimators span two orders of magnitude; signs are in Table 7. Panel B is the variance the correction costs, with ±1 across-replication standard deviation. Reading them together is the point: the cheapest weighting is not the one nearest the oracle, and the one nearest the oracle is not the same weighting in every law.

Four things follow from Table 7, and only the first is already in the documentation.

Neither ingredient suffices alone, and each fails completely in the other's regime. Against a naive bias of +0.389 SD under covariate-only recruitment, calibration alone leaves +0.383; against +0.302 under outcome-only recruitment, the participation model alone leaves +0.150. These are not partial corrections. The complementarity claim of `docs/studies.md` survives a harder test than the one that produced it.

The base model is not free. Under outcome-only recruitment, calibration from a uniform base is at +0.006 — indistinguishable from the oracle's +0.006 — while adding the participation model as a base moves it to −0.065, eleven times worse. The reason is structural, not statistical: $\mathbb{P}(S = 1 \mid X)$ does vary with X here, because X predicts Y and Y drives participation, so $1/\hat{P}(S \mid X)$ imports a covariate tilt that the outcome constraint has no way to remove. The package documentation says a base model on such a design "adds estimation noise". It adds bias, and the bias does not shrink with the sample. Where recruitment is believed to act through the diagnosis alone, the uniform base is the specification, not the lazy default.

The residuals are misspecification, not noise. The oracle sits within 0.003 SD of zero in every law, so the 0.036–0.083 left by `ipw+cal` in the three mixed laws is the part of the density ratio outside the tilt family, measured rather than argued. Note that even the additive law leaves a residual: a participation logit that is additive in X and Y does not have a log density ratio additive in X and Y, so "additive selection" is already slightly outside base-plus-marginal.

Augmentation is the most robust deployable choice here, within limits. `aipw` has the lowest or near-lowest RMSE in four of the five laws and halves the interaction law's bias, −0.083 to +0.033. It is not a repair for a misspecified tilt in general — under severity-dependent recruitment it is no better than the weights it augments — and it buys its accuracy with an outcome model that must itself be right.

One diagnostic result deserves separate statement, because it qualifies a recommendation the README makes without qualification. The worst held-out standardized mean difference *did* detect the base-model harm under outcome-only recruitment, 0.146 for `ipw+cal` against 0.034 for `cal`, ranking them as their biases do. Under additive recruitment it ranked them backwards: 0.024 for `ipw`, whose bias is +0.120, against 0.106 for `ipw+cal`, whose bias is −0.036. A held-out margin is an alarm — a large discrepancy is evidence of misspecification — and it is not a model-selection criterion. Choosing between two weightings by comparing their held-out balance can select the worse one.

9.2.2 How much register information is worth supplying (B2)

Table 8: B2. A ladder of register inputs against a fixed recruitment law with three outcome-side channels, 40 replications. † the co-occurrence moment is a constraint at that rung and matches by construction.

Register information	Bias	RMSE	ESS	worst |SMD|	comorbidity
none	+0.125	0.129	1756	0.024	+0.0188
K(Y1)	-0.037	0.058	1500	0.111	+0.0044
K(Y1), K(Y2)	-0.036	0.057	1501	0.110	+0.0037
+ co-occurrence	-0.036	0.057	1500	0.110	-0.0000†
per stratum	-0.026	0.052	1518	0.113	+0.0044
oracle	+0.001	0.023	1523	0.036	-0.0000

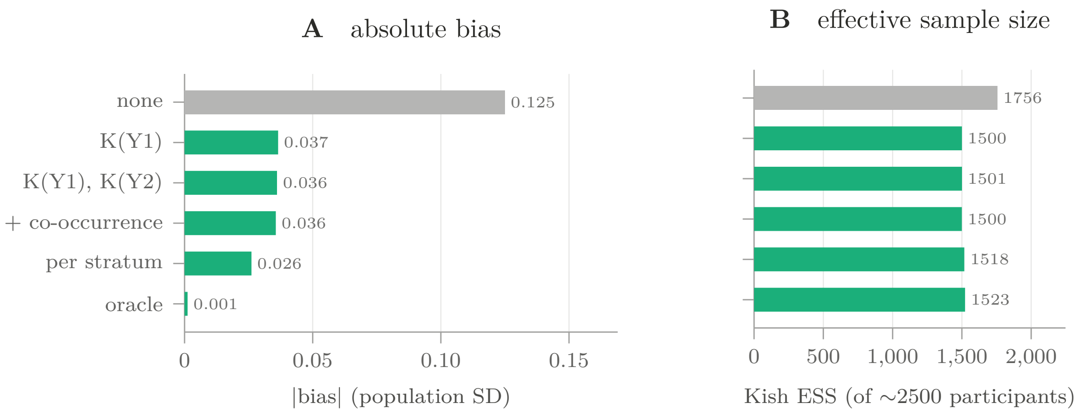

Figure 4: B2. What each additional register quantity buys (A) and what it costs (B). The oracle row is shaded. The first prevalence does nearly all of the work; the rungs above it are close to free in effective sample size, which is why the reason to stop adding constraints is support (§9.2.6) rather than variance.

The first prevalence takes the bias from +0.125 to −0.037. A second prevalence and a co-occurrence target then buy almost nothing on this estimand (−0.036 both), even though the co-occurrence constraint drives its own moment from +0.019 to zero — a clean instance of the general rule that a constraint fixes what it constrains and its neighbourhood, and nothing further. Per-stratum prevalences are the one rung that helps, −0.026.

The effective sample size is nearly flat across the ladder, 1500–1520 against 1756 for the participation model alone. Constraints in this design are cheap; almost the whole variance cost is paid by the first one. That is a useful practical fact and a dangerous one: cheapness is not a reason to add constraints, because each one still has to be a quantity the register actually knows, and B3 measures what happens when it does not.

9.2.3 A mis-stated register prevalence (B3)

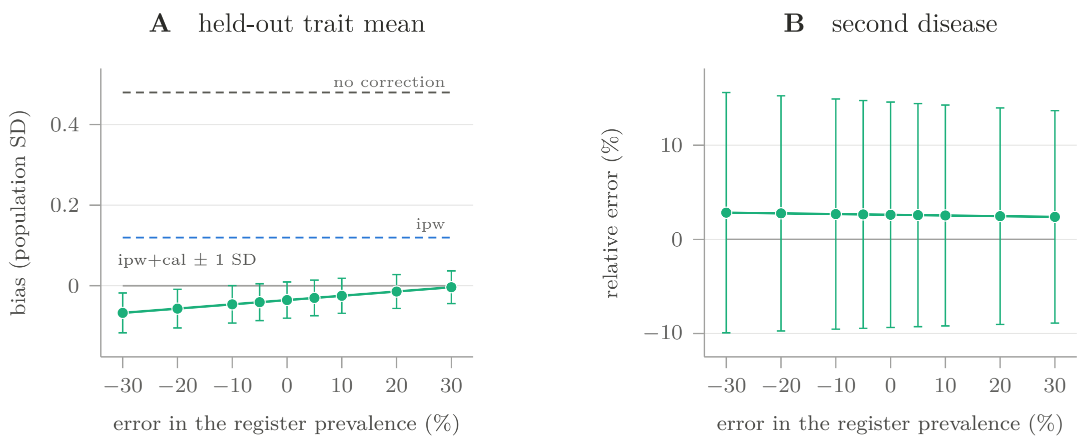

Figure 5: B3. The anchored prevalence is scaled by $1 + \delta$ and everything else held fixed, 40 replications. Panel A: the held-out trait, against horizontal reference lines for the two weightings that do not depend on δ. Panel B: the second disease, never anchored. Shaded whiskers are ±1 across-replication standard deviation. The anchored margin itself tracks δ exactly and is not plotted.

Over a ±30% sweep the held-out trait bias moves from -0.067 to -0.004 SD — about 0.011 SD per ten percentage points of relative register error, against a naive bias of $+0.479$. The unanchored disease moves from $+2.83\%$ to $+2.39\%$ relative error, and the effective sample size from 1444 to 1585. On this design the method is robust to the size of error a register plausibly carries: a diagnostic threshold that has drifted by a tenth costs roughly a fortieth of the bias it removes.

The sweep also shows why the recommendation is to *report* it rather than to read a minimum off it. The bias is smallest at $\delta = +30\%$, not at $\delta = 0$, because a positive target error happens to offset the estimator's own misspecification. A sensitivity sweep bounds an estimate over a range of register values. It cannot locate the truth inside that range, and the value of δ that minimizes the error is not an estimate of the register's error.

9.2.4 Case count and case mix (B4)

Table 9: B4. Three recruitment mechanisms against three things a calibration can be anchored to, 40 replications. Case mix is the error in mean liability among cases; severe share is the error in the fraction of cases the register would grade severe; within stratum is the worst error in a within-stratum prevalence. † constrained in that arm, and therefore an identity rather than a result.

Weighting	case mix	severe share	within stratum	trait bias
Recruitment: pooled only				
no correction	-0.0023	-0.0031	0.2741	+0.4793
ipw	-0.0284	-0.0253	0.2458	+0.1196
ipw+cal	-0.0284	-0.0253	0.0173	-0.0355
ipw+cal, strata	-0.0231	-0.0199	0.0000†	-0.0256
ipw+cal, severity	-0.0104	-0.0000†	0.0179	-0.0345
oracle $1/\pi$	-0.0043	-0.0032	0.0137	+0.0017
Recruitment: stratum-differential				
no correction	+0.0235	+0.0213	0.4643	+0.5475
ipw	-0.0150	-0.0135	0.4482	+0.1250
ipw+cal	-0.0150	-0.0135	0.1687	-0.0439
ipw+cal, strata	-0.0269	-0.0251	0.0000†	-0.0158
ipw+cal, severity	-0.0054	-0.0000†	0.1695	-0.0434
oracle $1/\pi$	-0.0035	-0.0040	0.0114	+0.0016
Recruitment: within-case severity				
no correction	+0.0839	+0.0758	0.3504	+0.5472
ipw	+0.0555	+0.0528	0.3079	+0.1576
ipw+cal	+0.0555	+0.0528	0.0258	-0.0404
ipw+cal, strata	+0.0627	+0.0593	0.0000†	-0.0263
ipw+cal, severity	+0.0156	-0.0000†	0.0238	-0.0422
oracle $1/\pi$	-0.0014	-0.0005	0.0123	+0.0023

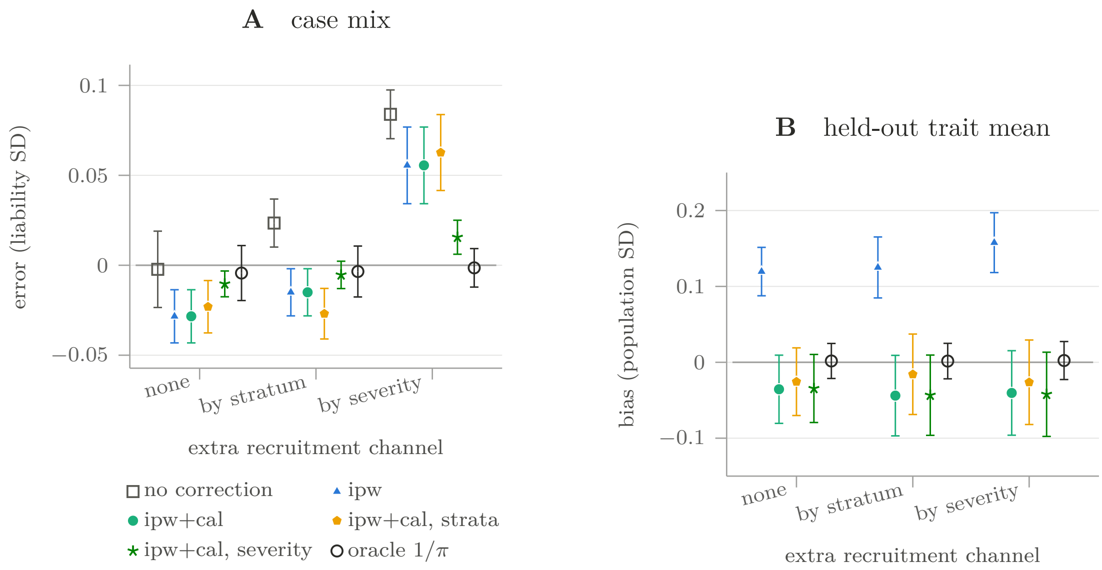

Figure 6: B4. Only quantities that are held out in every arm are plotted: the case mix (A) and the trait mean (B). The identities are in Table 9, marked. Whiskers are ±1 across-replication standard deviation.

The README's third recommendation — calibrate within strata, "usually the single most important step for psychiatric cohorts" — has had no number behind it. It now has one, and it comes with a correction to how the advice is stated.

Under stratum-differential ascertainment, a pooled calibration leaves the worst within-stratum prevalence in error by 0.169: larger than the population prevalence being estimated. Stratifying removes that by construction, and — the part that is evidence rather than algebra — improves the held-out trait from −0.044 to −0.016, most of the remaining distance to the oracle's +0.002.

Under within-case severity selection, stratifying on sex and birth year does *not* help: the case-mix error is +0.056 pooled and +0.063 stratified, against an oracle at −0.001. What helps is anchoring the register's mild- and severe-case prevalences separately, which brings it to +0.016. The general statement is therefore not "stratify" but *stratify along the axis recruitment acts on*, and which axis that is, is a claim about the recruitment mechanism that the calibration cannot supply. A cohort whose clinicians recruited through hospital contact should anchor severity; a cohort with uneven regional coverage should anchor region; anchoring the wrong one costs effective sample size and buys nothing.

Table 10: B5. Coverage of a nominal 95% interval for the held-out trait mean, 300 replications, 400 bootstrap resamples. Monte Carlo is ±1.96 binomial standard errors on the coverage. Width and bias are in population SD units.

Interval	coverage	Monte Carlo	width	bias
Specification: correct				
fixed-weight	0.960	±0.022	0.087	+0.0004
calibration-aware	0.953	±0.024	0.085	
bootstrap	0.940	±0.027	0.084	
Specification: base + margin				
fixed-weight	0.653	±0.054	0.110	-0.0359
calibration-aware	0.643	±0.054	0.105	
bootstrap	0.640	±0.054	0.104	
Specification: misspecified				
fixed-weight	0.373	±0.055	0.130	-0.0826
calibration-aware	0.357	±0.054	0.122	
bootstrap	0.343	±0.054	0.120	

9.2.5 Do the intervals cover? (B5)

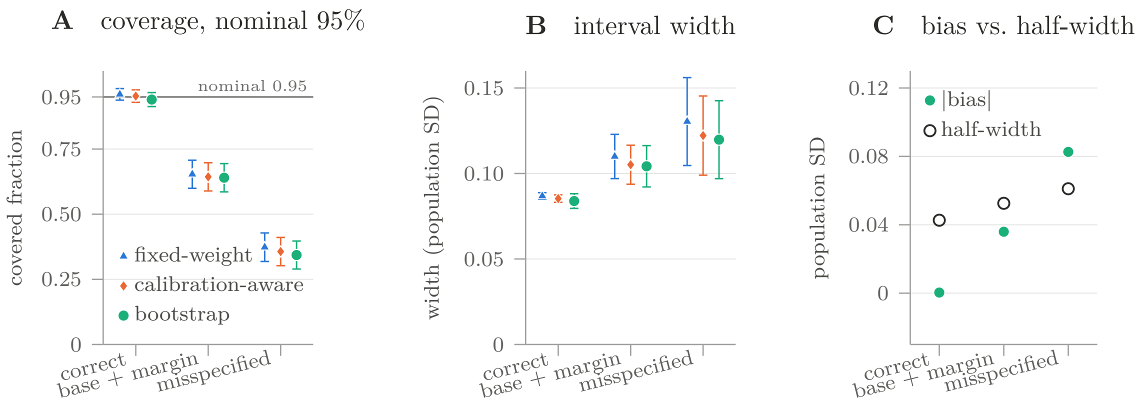

Figure 7: B5. Coverage against nominal (A), interval width (B), and the bias beside the half-width it would have to fit inside (C; open circles are the calibration-aware half-width). The three procedures are close together in every regime and the regimes are far apart, which is the finding: what decides coverage here is the specification, not the variance formula.

This is the item the previous edition of this note listed as absent from the evidence base, and the answer is sharp. Where the tilt family contains the truth — outcome-only recruitment, uniform base, calibrated on the outcome — all three procedures are at nominal: 0.960, 0.953 and 0.940, each within Monte Carlo error of 0.95. The variance machinery is correct.

Where it does not, coverage collapses: 0.64 under the ordinary applied combination of a fitted base and a marginal constraint, and 0.34–0.37 when recruitment carries a covariate-by-outcome interaction. The interval widths across those three regimes are 0.085, 0.105 and 0.122 SD — they barely move. The collapse is entirely bias. The calibration-aware half-widths are 0.043, 0.053 and 0.061 SD, so the biases are 1%, 68% and 135% of the half-width they would have to fit inside; an estimator displaced by more than its own half-width cannot cover at any nominal level.

The practical statement is that these intervals quantify sampling variability and say nothing whatever about the density-ratio specification, so a reported interval is conditional on an assumption it cannot test. That is not a defect of the estimator — no variance formula can cover a bias — but it does mean that an interval from `i3pw` should be reported alongside the held-out diagnostic of §9.2.4 and the sensitivity sweep of B3, never alone.

Two limits of this particular result. The estimand is nearly orthogonal to the single constraint, so the fixed-weight and calibration-aware formulas nearly coincide here; their documented difference is on the constrained margin itself, where the fixed-weight standard error is positive for a quantity the calibration reproduces exactly. And the bootstrap resamples participants without refitting the participation model, so it covers the calibration's variability and not the base model's.

9.2.6 How rare the disease can be (B6)

Table 11: B6. A prevalence sweep at a fixed participant count, 30 replications, 200 bootstrap resamples, population 30,000. "Solve fails" is the share of replications whose calibration could not reach the target; "discarded" is the share of bootstrap resamples that could not, conditional on the original solve succeeding. The over-recruited arm's solve never failed.

	cases over-recruited		cases under-recruited		
K	cases	discarded	cases	solve fails	discarded
0.2	1274	0.001	123	0.00	0.002
0.1	763	0.000	54	0.00	0.000
0.05	422	0.000	26	0.00	0.000
0.02	179	0.000	11	0.00	0.001
0.01	89	0.000	6	0.00	0.034
0.005	45	0.000	3	0.07	0.102
0.002	18	0.000	1	0.27	0.274
0.001	9	0.007	1	0.43	0.331

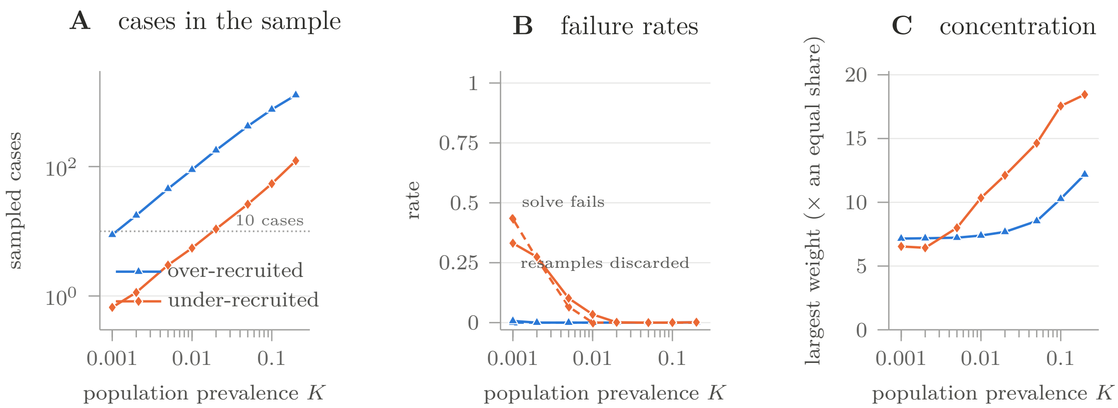

Figure 8: B6. Sampled cases (A), the two failure rates (B; dashed is the solve, solid is the bootstrap), and weight concentration (C), against population prevalence on a logarithmic axis. The direction of the disease channel decides which failure arrives: cases that participate *less* are the dangerous case, and the realistic one for severe illness.

The direction matters more than the rarity. Where cases are over-recruited the calibration is downweighting them, and the solve survives to nine sampled cases with a bootstrap discard rate under 1%. Where cases are under-recruited — the realistic direction for severe psychiatric illness, dementia, or anything that makes answering an invitation harder — the sample must carry the population's disease on few shoulders, and the failures arrive in a fixed order:

- at roughly five sampled cases ($K = 0.01$) the point estimate still solves in every replication, and 3.4% of bootstrap resamples are already being discarded;
- at three ($K = 0.005$) the solve fails in 7% of replications and 10% of resamples are discarded;
- at one ($K = 0.002$) the solve fails in 27% and 27% of resamples are discarded.

The ordering is the warning. The interval degrades roughly an order of magnitude in prevalence before the point estimate does, and it degrades silently: discarded resamples are exactly the ones poorest in cases, so what remains is a truncated sampling distribution and a percentile interval that is too narrow. A calibration that solves cleanly can still be carrying an interval that does not mean what it says, and `BootstrapResult.failure_rate` is the only thing that reveals it. Weight concentration is also worse in this direction, peaking at 18.4 times an equal share against 12.2 when cases are over-recruited.

9.2.7 The ridge (B7)

Table 12: B7. The ridge on the tilt across four orders of magnitude, under three specifications, 40 replications. The ridge column "0 (exact)" is calibration to the target exactly.

Ridge	0 (exact)	0.001	0.01	0.03	0.1	0.3	1
Specification: correct							
|bias|	0.0058	0.0076	0.0223	0.0483	0.1054	0.1784	0.2470
RMSE	0.0222	0.0227	0.0308	0.0528	0.1081	0.1808	0.2496
ESS	1906	1912	1957	2036	2203	2378	2482
Specification: base + margin							
|bias|	0.0355	0.0344	0.0252	0.0092	0.0244	0.0632	0.0959
RMSE	0.0569	0.0561	0.0501	0.0426	0.0454	0.0723	0.1013
ESS	1517	1519	1540	1574	1639	1701	1738
Specification: misspecified							
|bias|	0.0832	0.0820	0.0722	0.0551	0.0197	0.0199	0.0520
RMSE	0.1055	0.1044	0.0959	0.0819	0.0588	0.0534	0.0687
ESS	1238	1239	1254	1280	1331	1385	1425

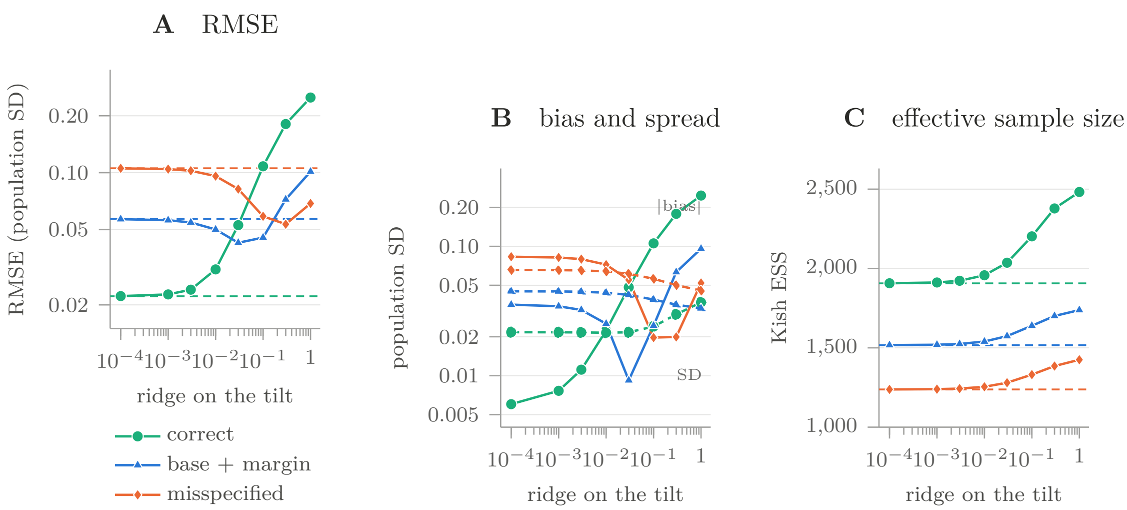

Figure 9: B7. RMSE against the ridge (A), with each specification's exact solve as a dashed rule; the bias and the across-replication spread separately (B; solid is |bias|, dashed is the spread); and the effective sample size the ridge buys (C). Under a correct specification the exact solve is the right answer. Under misspecification it is not.

The natural reading of `shrinkage=` — exactness traded for stability — is right in one of the three arms and wrong in the other two.

Where the tilt family contains the truth, the exact solve is the minimum and every unit of ridge is a pure loss: RMSE climbs from 0.0222 to 0.2496, and it climbs because the *bias* grows from 0.006 to 0.247 while the across-replication spread barely moves, 0.0217 to 0.0369. There is no variance being bought. Under a correct specification the constraint is not costing precision, so relaxing it purchases nothing and gives up consistency.

Under both misspecified arms the minimum is at a positive ridge, and by a wide margin: 0.0426 at $\rho = 0.03$ against 0.0569 exact for the applied combination, and 0.0534 at $\rho = 0.3$ against 0.1055 exact for the interaction law — a halving of RMSE. Again the movement is in the bias ($0.083 \to 0.020$), not the spread. The exact constraint is being met in a direction the tilt family cannot support, and relaxing it lets the weights stay nearer a base that is closer to the truth.

The two facts together are why this is not a tuning recipe. The ridge that halves the error under the interaction law, $\rho = 0.3$, makes the correctly specified arm eight times worse; and the degree of misspecification, which is what would decide the choice, is exactly what a real analysis cannot observe. A ridge is a defensible response to a weighting that will not solve or whose weights concentrate. It is not a free accuracy improvement, and a sweep over ρ on real data has no held-out quantity to select against.

One diagnostic warning falls out of the same table, and it repeats the pattern of §9.2, B1. In the correctly specified arm the effective sample size *rises* monotonically with the ridge, 1906 to 2482, while the estimate gets eleven times worse. A healthier-looking ESS is not evidence of a better weighting; here it is produced by the same mechanism that destroys the estimate.

9.2.8 What the suite still does not establish

The suite is simulation. Everything above is a statement about a data-generating process chosen by the author of the method, and the honest reading of it is that the estimator behaves as the theory says it should in cases where the theory can be checked — not that a biobank will. Specifically:

- **No real cohort.** There is still no empirical demonstration of bias reduction in iPSYCH, UK Biobank, or anywhere else. Nothing in this section changes the milestone named in §10.
- **One family of recruitment laws.** The channels are switched on inside a single Bernoulli-logistic model over Gaussian covariates and a threshold liability. A real recruitment process need not be in that family at all, and the misspecified arms explore departures the author thought of.
- **The register is exact except in B3.** Elsewhere the targets are the realized population values, which no register supplies.
- **The participation model is well specified as a model class.** It is a LASSO logistic on the true frame covariates. A real frame is coarser than the variables that drive participation, and none of these numbers price that in.
- **Coverage is conditional on the design.** The bootstrap holds the base model fixed, and the estimand is nearly orthogonal to the constraint; both are stated at §9.2.5 and both would move the numbers.
- **Optimal tuning is not transferable.** The ridge that minimizes RMSE in B7 depends on the degree of misspecification, which is exactly the thing a real analysis does not know.

9.3 Recommended validation matrix

This is the matrix §9.2 was written against. The final column records how far the suite gets, so that the gap between what a methods paper needs and what this repository currently has is a table entry rather than a judgement.

Table 13: Minimum simulation and empirical validation matrix for a methods paper, and what the 0.3.1 benchmark suite covers. “Partly” means the axis is exercised but not at every level the first column asks for.

Axis	Levels	Primary outputs	Covered
Selection law	X-only; Y-only; additive $X+Y$; $X \times Y$; within-case severity; genotype-plus-outcome.	Bias, RMSE, interval coverage, ESS, maximum weight.	B1, B5. All but the genotype-plus-outcome level; collider bias on an effect size is left to the example scripts.
Anchor information	Correct pooled prevalence; stratum-specific prevalence; joint comorbidity; perturbed targets; partial anchors.	Anchor residual and transfer error on held-out outcomes and margins.	B2, B3, B4. All five levels.
Support	Common, rare, and extremely rare disease; sparse ancestry/sex/age cells.	Failure probability, convergence, weight concentration, bootstrap discard rate.	B6. Prevalence down to 2×10^{-3} in both recruitment directions; sparse stratum cells not swept.
Base model	Correct, misspecified, regularized, cross-fitted, and absent.	Incremental value of the base versus the prevalence tilt.	Partly: B1 has absent and present, B7 has regularized, B1 has a cross-fitted outcome model. A deliberately misspecified base class is not tested.
Comparator	Unweighted, covariate IPW, exact calibration, ridge calibration, oracle full-population law in simulation.	Bias–variance trade-off on the same target estimand.	B1, B7. All five, on one held-out estimand throughout.
Real data	Prespecified fit margins and separate held-out register margins or recruitment wave.	External discrepancies with uncertainty, not merely anchored matches.	Not covered. This remains the open milestone.

10 Limits and conclusion

The estimator is prevalence-calibrated density-ratio weighting, not unrestricted inference of individual inclusion probabilities. It is true full-population IPW when equation (19) holds. Otherwise it is the information projection of the base weights onto the supplied moments.

For statistical genetics the method is a generalization of case–control ascertainment correction: a frame-based participation model, several diagnoses or stratum-specific margins, and an inspectable tilt. A marginal prevalence cannot identify a joint law. Exact anchors are tautologies. Weak support creates extreme weights. Downstream AIPW does not repair a misspecified tilt family.

The numeric evidence in §9 has two layers. The 0.3.0 freeze in `report/validation_results.tsv` supports the narrow messages already in the documentation: transfer across unanchored diseases is weak, the covariate base matters when selection has a covariate channel, and anchored zeros are identities. The 0.3.1 benchmark suite in `report/benchmark_results.tsv` adds four statements that change how the estimator should be used rather than how it should be described.

1. **The base weights are part of the specification, not a free improvement.** Where recruitment acts through the diagnosis alone, a covariate participation model used as a base makes the estimate eleven times worse than a uniform base, because it imports a covariate tilt the outcome constraint cannot remove (§9.2, B1).
2. **Reported intervals are conditional on the tilt family.** They are at nominal when it contains the truth and at 0.64 under the ordinary combination of a fitted base and a marginal constraint, with almost no change in width: the loss is bias, and no variance formula covers it (B5).
3. **Stratify along the axis recruitment acts on.** Demographic strata do not repair severity-dependent recruitment; separate severity-grade prevalences do. Which axis is right is a claim about the mechanism, and the calibration cannot supply it (B4).
4. **The interval fails before the point estimate does.** Under the realistic direction — cases participating less — bootstrap replicates start being discarded around five sampled cases, roughly an order of magnitude in prevalence before the solve itself fails, and the discards are silent (B6).

None of this establishes bias reduction in a real cohort. The next scientific milestone is unchanged: a prespecified held-out-register or later-wave study.

A Software and remaining engineering limits

A.1 Architecture and public contract

The source declares version 0.3.1 and an alpha development status. Its design has a sensible separation between a real-cohort interface and simulation machinery. The example freeze cited in §9 is still 0.3.0; the benchmark artifact is 0.3.1 and is regenerated by `python -m benchmarks.run_all`.

A.2 Scientific contracts that must remain explicit

A.3 Computational properties

Let n be the number of participants with positive base weight, k the number of constraints, p the number of frame covariates, I optimizer iterations, and B bootstrap replicates.

For the core calibration, k is normally small and the implementation is reasonable. The main practical failure is statistical rather than computational: adding many sparse stratum-by-disease constraints can exhaust support before it exhausts RAM. The liability module is the exception; materializing n^2 pairwise similarities is untenable at modern biobank scale.

A.4 Remaining release gaps

Table 14: Main software components in the reviewed `i3pw` worktree.

Component	Public role	Review
`calibrate` and `CalibrationFit`	Real-cohort entry point using participant outcomes, external targets, optional base weights, strata, joint moments, and held-out margins.	The correct conceptual front door; it avoids requiring simulated ground truth.
`entropy_balance` and `apply_tilt`	Core dual solve and transfer of a fitted tilt to new rows.	Small convex problem, log-scale stabilization, explicit diagnostics, and preservation of zero-weight support.
`lasso_propensity` and `lasso_ipw`	Covariate-only participation baseline.	Useful comparator, but a real application needs frame-level participation labels and compatible covariates for participants and nonparticipants.
`calibration_ipw` and `monte_carlo`	Simulation wrappers with known population truth.	Appropriate for method development, not the interface to advertise as real-data inference.
`balance` and `uncertainty`	Held-out discrepancies, effective sample size, linearization, bootstrap, and target sensitivity.	Good instincts; target uncertainty, clustering, and a formal specification test remain user responsibilities.
`aipw`	Outcome-regression augmentation for a participant-only trait when X is known for the population frame.	Cross-fitting is available; robustness claims still require the usual missing-data conditions.
`liability`	Threshold-model simulations and weighted pairwise moment estimators.	Scientifically relevant, but the explicit $n \times n$ similarity matrix does not scale to biobank sample sizes.

Table 15: High-impact semantic and numerical contracts in the corrected design.

Issue	Correct contract	Consequence
Meaning of the tilt	λ belongs to the log population-to-participant density ratio, as in equations (9)–(10).	Do not report it as the outcome coefficient in a participation logit.
Target of base weights	$1/\widehat{\pi}$ transports participants to the full population; $(1-\widehat{\pi})/\widehat{\pi}$ transports them to nonparticipants.	The inverse scheme is the full-population default; an odds option needs an explicit different target.
Simulation oracle	A full-population oracle is the complete simulated test-fold mean, unavailable in real data.	An inverse-odds mixture is not a full-population oracle and should not be labelled one.
Ridge calibration	The achieved residual equals $-\rho\widehat{\lambda}_\rho$, and the influence projection contains the same ρ.	Only exact $\rho = 0$ anchors have zero conditional standard error.
Zero base weights	Zero-weight rows are excluded from the exponential shift and remain zero.	An extreme feature value outside the positive support must not destabilize all weights.
Held-out balance	The SMD is a heuristic discrepancy using an unused margin.	It may be called a falsification diagnostic, not a formal J-test.
Synthetic selection	The generator uses Bernoulli logistic participation with coefficients chosen so expected sample count and outcome totals meet configured targets.	Realized sample size and prevalences are random; the mechanism is not fixed-size probability-proportional-to-size sampling.

Table 16: Leading computational costs. These are algorithmic orders, not timing benchmarks.

Operation	Approximate time	Approximate memory	Comment
Entropy dual	$O(Ink)$	$O(nk)$ for features	L-BFGS avoids forming a dense $k \times k$ Hessian.
Stratified design	$O(nk)$	$O(nk)$	k can be roughly strata times outcomes, plus share and interaction constraints.
LASSO participation model	Solver- and CV-dependent	$O(np)$ before expansion	Pairwise interactions increase the design toward $O(np^2)$.
Calibration bootstrap	$O(BInk)$ plus optional base refits	One replicate plus stored output	Parallelization is natural, but rare anchors can cause infeasible replicates.
Liability pairwise moment	$O(n^2p)$ to form $XX^\intercal$	$O(n^2)$	This is the principal barrier to biobank-scale use. Blocked matrix products or summary-statistic identities are needed.
Synthetic data generator	Repeated $O(Nq)$ selection objectives, plus covariance work	$O(Np + Nq + p^2)$	Useful for controlled experiments; not normally a production bottleneck.

Table 17: Important gaps that remain after correcting the scientific semantics.

Gap	Consequence	Recommended next step
Real-data resampling interface	The bootstrap and prevalence-sensitivity helpers take the simulation `Dataset`; users of `calibrate` must assemble equivalent refits themselves.	Provide a callback- or array-based bootstrap that refits the base, tilt, and estimand without simulated ground truth.
External-target and cluster variance	`CalibrationFit.mean` treats external targets as fixed and independent units as the sampling units.	Accept target covariance and cluster identifiers, or document a supported cluster-bootstrap recipe.
Liability scaling	The explicit $n \times n$ similarity matrix is incompatible with biobank-scale n.	Use blocked accumulations, randomized trace methods, or algebraic sufficient statistics that avoid storing pairwise matrices.
Rendered documentation	The corrected distributions include the Markdown theory, examples, LaTeX report, and citation record, but there is no rendered versioned site or link checker.	Publish versioned HTML/PDF documentation and check external and internal links in CI.
Frozen environment and automated evidence	A citation file, structured post-correction results, and a local environment manifest are now present, and CI smoke-tests one packaged example; there is no cross-platform dependency lock, replicated-validation job, or comparison of outputs with the frozen table.	Record a locked environment and add a reproducible benchmark job with explicit tolerances and retained logs.

B Analysis protocol

A defensible analysis can be organized in nine steps.

1. **Define T.** State geography, calendar period, age range, ancestry scope, survival condition, and disease definition. A prevalence from a different population is not a magic number; it is a different estimand.
2. **Define the target quantity.** A prevalence, population mean, regression coefficient, and liability-scale variance component need different estimating equations. Weighting is not the estimand.
3. **Build the base correction.** Fit participation using variables observed compatibly in the cohort and its sampling frame. For the full population use $1/\widehat{\pi}(X)$, not inverse odds. Separate training and analysis folds when a flexible model is used.
4. **Prespecify g.** Begin with scientifically justified prevalence margins. Add strata or interactions only when population targets and participant support exist. Reserve some credible margins for validation.
5. **Fit and inspect.** Report convergence, residuals, tilt coefficients, support counts, ESS, range, quantiles, and top-weight mass. For ridge, report ρ and achieved rather than nominal margins.
6. **Try to refute the weighting.** Evaluate held-out margins and a transferred tilt on a later fold or wave. Do not call an SMD threshold a p-value.
7. **Estimate uncertainty at the right level.** Use the penalized calibration influence adjustment corresponding to the fitted ρ, or re-fit the entire pipeline in a bootstrap. Respect family clusters and quantify uncertainty in external targets.
8. **Stress the assumptions.** Perturb prevalence, alter trimming or ridge, coarsen strata, and examine plausible within-case selection. Trimming after exact calibration breaks the constraint and must be followed by reporting the new residual.
9. **Report weighted and unweighted analyses.** Large changes deserve explanation; small changes can reflect either little bias or an uninformative weight. Neither is established by the point estimate alone.

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Table 18: Items that should accompany a published `i3pw` analysis.

Item	Minimum disclosure
Target population	Eligibility, time, place, ancestry handling, outcome definition, and source of every external margin.
Base model	Training frame, predictors, missing-data handling, validation, clipping, and whether weights are $1/\pi$ or inverse odds.
Calibration model	All g columns and targets, exact or ridge objective, ρ, strata, joint moments, and any post-fit trimming.
Support and concentration	Cases and controls per cell, ESS, weight quantiles, maximum, and top-1% mass.
Validation	Which margins were held out before fitting, their discrepancies, and whether the tilt was transferred to independent data.
Inference	Influence or bootstrap procedure, refitted stages, clustering unit, failed replicates, and treatment of external-target uncertainty.
Sensitivity	Plausible target perturbations and departures from within-class exchangeability.
Reproducibility	Software version or commit, random seeds, environment, frozen tables, and code that recreates every reported number.

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